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|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a)(i)  | number of (different) species (of plant) / species richness ;<br>abundance / number / frequency / population / density, of each species (of plant) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>2</b> |
| 1(a)(ii) | ref. to trampling / trampled versus non-trampled ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>1</b> |
| 1(b)     | <p><i>any <b>eight</b> from:</i></p> <ol style="list-style-type: none"> <li>1 use, quadrat(s) / a (named type of) transect ;</li> <li>2 <i>idea of</i> sampling in the trampled and untrampled area ;</li> <li>3 ref. to, <b>method</b> of, random placement of quadrats / systematic sampling along the transect ;</li> <li>4 ref. to use of, same / stated, size (frame / point), quadrat ;</li> <li>5 method of identifying species ;</li> <li>6 ref. to a method of estimating abundance ;</li> <li>7 <i>idea of</i> taking care to note low growing / small / easy to miss, species ;</li> <li>8 <i>idea of</i> sampling at least 3 separate points in the open land and on the shortcut path / 3 different line transects ;</li> <li>9 repeat (whole investigation) at different times of, year / seasons ;</li> <li>10 calculate / use, Simpson's index (of diversity) ;</li> <li>11 plants causing irritation / allergy <b>and</b> wearing, gloves / protective clothing<br/> <b>or</b><br/> pollen causing, hay fever / allergy, <b>and</b> wearing mask or taking medication<br/> <b>or</b><br/> ref. to environmental hazard <b>related to</b> its risk <b>and</b> wearing, suitable shoes / protective clothing or suitable mitigation ;</li> </ol> | <b>8</b> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(c)(i)   | <p><i>support:</i><br/>(all) the times, for path / trampled area, are, longer / higher / slower (than untrampled) / <b>ora</b> ;</p> <p><i>do not support:</i><br/><i>idea of</i> no information about soil air spaces ;</p>                                                                                                                                                                                                                                                           | 2     |
| 1(c)(ii)  | <p><math>S_M = 2.8</math> ;</p> <p>95% confidence interval = 68.1 / 68.2, to 79.5 / 79.4 ;</p>                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| 1(c)(iii) | <p><i>any two from:</i></p> <p>1 95%, probability / confidence that / chance that, the, true / population / parametric / actual, mean, lies within these limits ;</p> <p>2 the difference between means is (likely to be) significant (at <math>p &lt; 0.05</math> because there is no overlap between 95% CI for untrampled and trampled) ;</p> <p>3 <i>idea that</i> the narrower CI (in the untrampled area), the closer the calculated mean is to the true mean / <b>ora</b> ;</p> | 2     |
| 1(d)(i)   | <p><i>any two from:</i></p> <p>1 <i>idea of</i> using a grid and tracing outline / using transparent grid over leaf ;</p> <p>2 <i>idea of</i> systematic counting of squares fully and more than half covered ;</p> <p>3+4 AVP ; ;</p>                                                                                                                                                                                                                                                 | 2     |
| 1(d)(ii)  | <p>there is a negative correlation<br/><b>or</b><br/>as time taken for (10 cm<sup>3</sup>) of water to drain through decreases, leaf surface area increases / <b>ora</b> ;</p>                                                                                                                                                                                                                                                                                                         | 1     |
| 1(d)(iii) | <p>1 working at, <math>p = 0.05</math> / <math>p &lt; 0.05</math> ;</p> <p>2 correlation / relationship, is, significant / not due to chance, as calculated <math>r_s</math> (0.455) is greater than (critical value of) 0.447 ;</p>                                                                                                                                                                                                                                                   | 2     |